

Project Title
(Pharma Search)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Pharma Search is an intelligent medicine search and recommendation system designed to improve medicine retrieval in pharmacies, hospitals, and healthcare organizations. The system enables users to search medicines using attributes such as medicine name, manufacturer, dosage, and category while providing complete medicine details and suitable substitute recommendations when a medicine is unavailable. To ensure fast and accurate retrieval from large medicine databases, the project applies advanced Data Structures and Algorithms, including string pattern matching techniques such as Knuth-Morris-Pratt (KMP) and Rabin-Karp, along with Dynamic Programming-based Edit Distance for fuzzy search and spelling correction. The application is implemented using Java with MySQL as the backend database. By reducing search time, minimizing errors caused by misspelled medicine names, and improving pharmacy workflow, Pharma Search demonstrates the practical application of advanced algorithms in solving real-world healthcare information retrieval problems.

Signature of the Guide